

ME 333
MECHANICS OF MATERIALS

Lecture 16: Unsymmetric Bending I

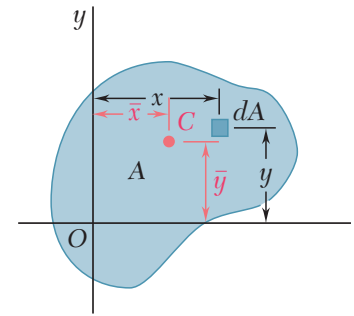
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Moments of Area

First Moment of Area :



$$\bar{x} = \frac{\int x dA}{\int dA} = \frac{Q_y}{A}; \quad \bar{y} = \frac{\int y dA}{\int dA} = \frac{Q_x}{A}$$

$$Q_y = A\bar{x};$$

$$Q_x = A\bar{y}$$

- If Q is zero then the centroid coincides with the origin of the coordinate axis.
- The first moment of area about the axis of symmetry is zero.

Ref Base Fig: Mechanics of Materials, Beer & Johnston

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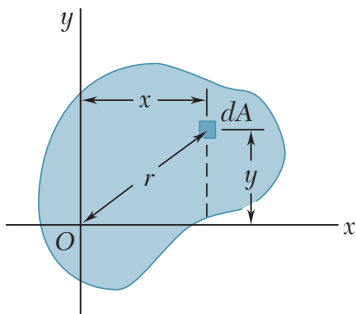
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Moments of Area

Second Moment of Area [Area Moment of Inertia]



$$I_{xx} = \int_A y^2 dA; \quad I_{yy} = \int_A x^2 dA;$$

$$J = \int_A r^2 dA = \int_A (x^2 + y^2) dA = I_{xx} + I_{yy}$$

- Moment of inertia represents the **resistance** offered by the body **to rotation**.
- In particular, the area moment of inertia signifies the **resistance** offered **to bending**.

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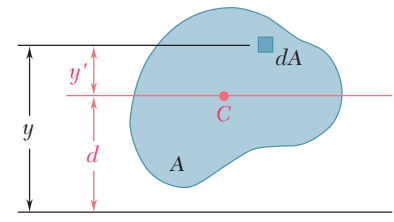
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Moments of Area

Parallel Axis Theorem



$$I_{xx} = \int_A y^2 dA = \int_A (d^2 + y'^2) dA = Ad^2 + I_{x'x'}$$

- It is a useful theorem in calculating the moment of inertia measures.
- In case of **complex geometries**, the geometry is **divided into sub-regions** which allow in calculating the moment of inertia for the entire geometry.

Ref Base Fig: Mechanics of Materials, Beer & Johnston

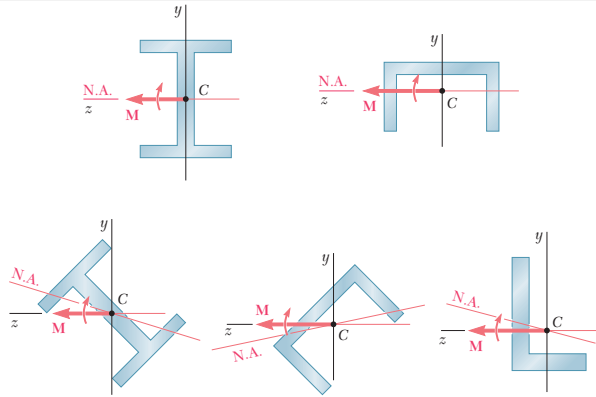
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Unsymmetric Bending of Beams



Q How are stresses calculated ?

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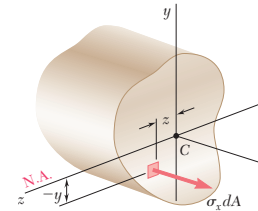
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Unsymmetric Bending of Beams

- NOTE : As long as we have at least one axis of symmetry we can easily solve unsymmetrical bending problems.
- Axis of symmetry coincides with principal centroidal axes and principal centroidal axes are perpendicular to each other



$$\sum F_x = \int_A \sigma_x dA = 0$$

$$\sum M_z = \int_A (-y\sigma_x) dA = M_z \quad (\text{resultant})$$

$$\sum M_y = \int_A z\sigma_x dA = M_y \quad (\text{resultant})$$

$$I_{yz} = \int_A yz dA = 0$$

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Unsymmetric Bending of Beams

Q What if the cross-section is not symmetric or direction of bending moment does not co-incide with the principal centroidal axes ?

- Step 1: We find the principal centroidal axis of the cross-section
- Step 2: We resolve M into components along the principal centroidal axes
- Step 3: We apply the superposition principle.

Q How do we do that ?

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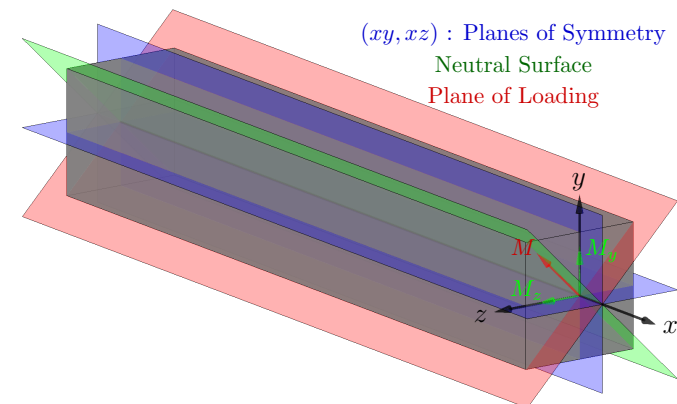
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Unsymmetric Bending of Beams

Example 1: Rectangular cross-section

Unsymmetric Bending



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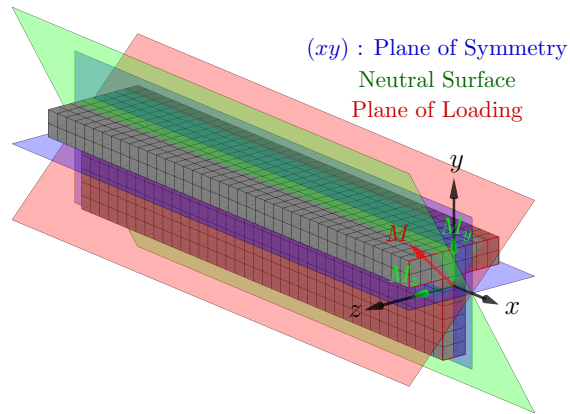
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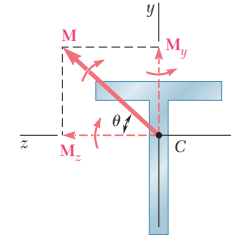
Unsymmetric Bending of Beams

Example 2: T-section Beam



Unsymmetric Bending of Beams

Example 2: T-section Beam (Contd ...)



Steps :

- Note that y and z are principal centroidal axes because y - axis is the axis of symmetry.
- Resolving components of M :

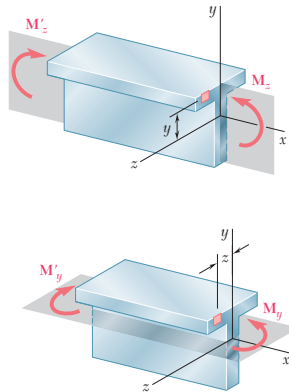
$$M_z = M \cos \theta; \quad M_y = M \sin \theta$$

$$\frac{M_y}{M_z} = \tan \theta$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

Unsymmetric Bending of Beams

Example 2: T-section Beam (Contd ...)



Steps :

- Moments about respective axes are:

$$\sigma_x = -\frac{M_z y}{I_{zz}}; \quad \sigma_x = \frac{M_y z}{I_{yy}}$$

- Applying superposition :

$$\sigma_x = -\frac{M_z y}{I_{zz}} + \frac{M_y z}{I_{yy}}$$

Ref Base Fig: Mechanics of Materials, Beer & Johnston

Unsymmetric Bending of Beams

Example 2: T-section Beam (Contd ...)

Q What about the Neutral Axis ?

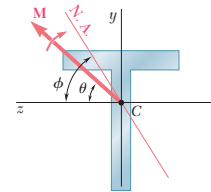
Steps :

- Neutral axis corresponds to the location where the normal stress is zero, therefore:

$$-\frac{M_z}{I_{zz}} y + \frac{M_y}{I_{yy}} z = 0$$

$$y = \left(\frac{I_{zz}}{I_{yy}} \tan \theta \right) z = (\tan \phi) z$$

- ϕ is the slope of N. A. w. r. to z -axis ($\tan \phi$)



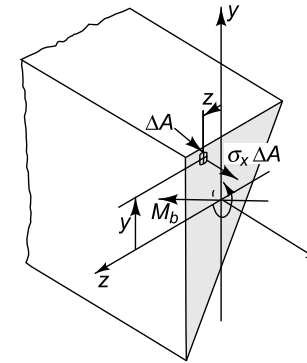
Ref Base Fig: Mechanics of Materials, Beer & Johnston

Unsymmetric Bending of Beams

Q What if the principal centroidal axes do not coincide and there is I_{xy} present?

Q How to handle cases where the beam has no planes of symmetry?

Unsymmetric Bending in Unsymmetric Beams

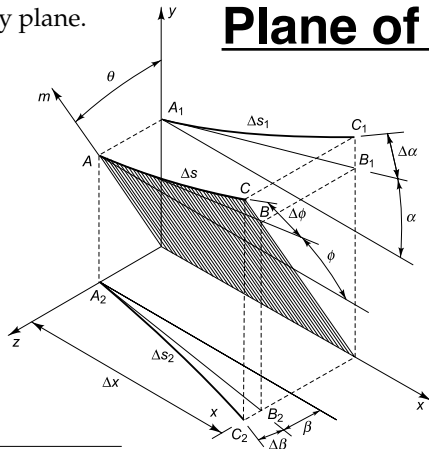


Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Unsymmetric Bending in Unsymmetric Beams

- $x - m$ is any arbitrary plane.

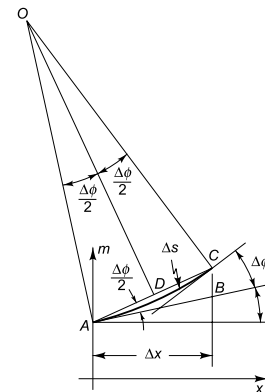
Plane of Loading (x-m)



Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

Unsymmetric Bending in Unsymmetric Beams

Curvature



Ref Base Figs: An Introduction to Mechanics of Solids, Crandall et. al.

- Curvature of arc AC is:

$$\frac{d\phi}{ds} = \lim_{\Delta s \rightarrow 0} \frac{\Delta\phi}{\Delta s}$$

- Length BC is then:

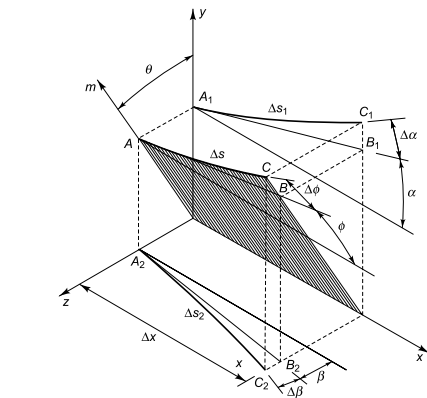
$$BC = \Delta x \frac{\Delta\phi}{2}$$

- Curvature can be defined in terms of BC as:

$$\frac{d\phi}{ds} = \lim_{\Delta x \rightarrow 0} \frac{2BC}{(\Delta x)^2}$$

Unsymmetric Bending in Unsymmetric Beams

Curvature



- Curvature of arcs A_1C_1 & A_2C_2 is:

$$\frac{d\alpha}{ds_1} = \lim_{\Delta x \rightarrow 0} \frac{2B_1C_1}{(\Delta x)^2}$$

$$\frac{d\beta}{ds_2} = \lim_{\Delta x \rightarrow 0} \frac{2B_2C_2}{(\Delta x)^2}$$

- Relating B_1C_1 & B_2C_2 to BC :

$$B_1C_1 = BC \cos \theta$$

$$B_2C_2 = BC \sin \theta$$

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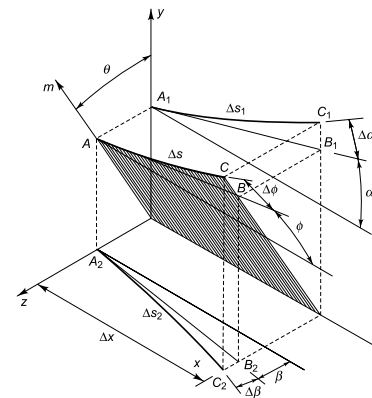
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Curvature



- Substituting with $d\phi/ds$ in the curvature relations:

$$\frac{d\alpha}{ds_1} = \frac{d\phi}{ds} \cos \theta$$

$$\frac{d\beta}{ds_2} = \frac{d\phi}{ds} \sin \theta$$

- Note : The curvatures of the projected curves are simply the components of the curvature of the original curve.

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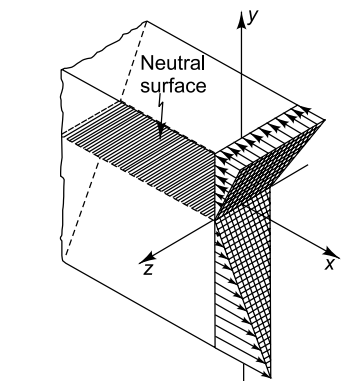
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Unsymmetric Bending in Unsymmetric Beams

Stress and strain descriptions



For moment about z-axis:

- Strain can be calculated based on geometric analysis as:

$$\epsilon_x = -\frac{y}{\rho_1} = -\frac{d\alpha}{ds_1} y$$

- Stress can be estimated using Hooke's law as:

$$\sigma_x = -E \frac{d\alpha}{ds_1} y$$

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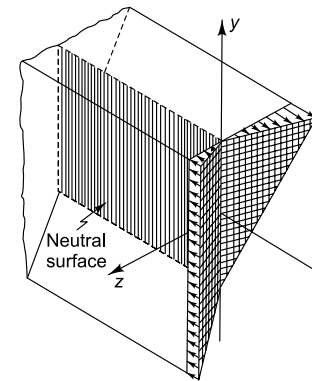
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Unsymmetric Bending in Unsymmetric Beams

Stress and strain descriptions



For moment about y-axis:

- Strain can be calculated based on geometric analysis as:

$$\epsilon_x = -\frac{z}{\rho_2} = -\frac{d\beta}{ds_2} z$$

- Stress can be estimated using Hooke's law as:

$$\sigma_x = -E \frac{d\beta}{ds_2} z$$

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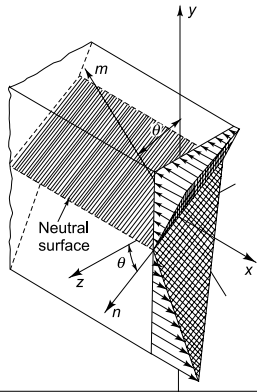
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Unsymmetric Bending in Unsymmetric Beams

Stress and strain descriptions



Using superposition :

- The total stress can be written as:

$$\sigma_x = -E \left(\frac{d\alpha}{ds_1} y + \frac{d\beta}{ds_2} z \right)$$

- Displacements and strains can be superposed, NOT Stresses. However, since there is a linear relationship between stress and strain and the material properties along the two directions is same - superposition is applicable.

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Unsymmetric Bending in Unsymmetric Beams

Moment equilibrium

- Moment about the z axis:

$$\begin{aligned} M_z &= - \int_A y \sigma_x dA = E \left(\frac{d\alpha}{ds_1} \int_A y^2 dA + \frac{d\beta}{ds_2} \int_A yz dA \right) \\ &= E \left(I_{zz} \frac{d\alpha}{ds_1} + I_{yz} \frac{d\beta}{ds_2} \right) = E \frac{d\phi}{ds} (I_{zz} \cos \theta + I_{yz} \sin \theta) \end{aligned}$$

- Moment about the y axis:

$$\begin{aligned} M_y &= \int_A z \sigma_x dA = -E \left(\frac{d\alpha}{ds_1} \int_A yz dA + \frac{d\beta}{ds_2} \int_A z^2 dA \right) \\ &= -E \left(I_{yz} \frac{d\alpha}{ds_1} + I_{yy} \frac{d\beta}{ds_2} \right) = -E \frac{d\phi}{ds} (I_{yz} \cos \theta + I_{yy} \sin \theta) \end{aligned}$$

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Unsymmetric Bending in Unsymmetric Beams

- Bending stress distribution:

$$\sigma_x = -E \left(\frac{d\alpha}{ds_1} y + \frac{d\beta}{ds_2} z \right) = -E \frac{d\phi}{ds} (y \cos \theta + z \sin \theta)$$

- Solving for curvature from moment equilibrium equations we can obtain the moment-curvature relationship as:

$$\begin{aligned} \frac{d\alpha}{ds_1} &= \frac{d\phi}{ds} \cos \theta = \frac{M_z I_{yy} + M_y I_{yz}}{E(I_{zz} I_{yy} - I_{yz}^2)} \\ \frac{d\beta}{ds_2} &= \frac{d\phi}{ds} \sin \theta = \frac{M_z I_{yz} + M_y I_{zz}}{E(I_{zz} I_{yy} - I_{yz}^2)} \end{aligned}$$

- Stress in terms of bending moments can be written as:

$$\sigma_x = - \frac{(y I_{yy} - z I_{yz}) M_z + (y I_{yz} - z I_{zz}) M_y}{(I_{zz} I_{yy} - I_{yz}^2)}$$

Ref : An Introduction to Mechanics of Solids, Crandall et. al.

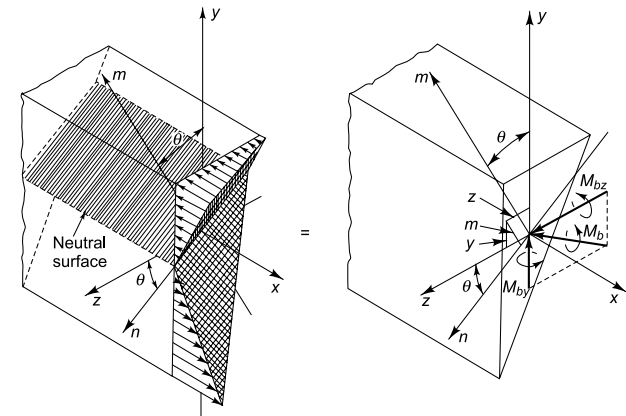
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Reading Assignment

Chapter 7 Crandall et. al.